

HfNbTaTiZr: supplementary methods and results

Fit diagnostics, processing sensitivity and complete records

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We analyze nine loading records containing 16009 stress-strain pairs. This supplement uses the article’s dataset and analysis configuration. Each composition contributes one atomic realization; the sensitivity ranges below measure changes in curve processing.

1 Initial-slope diagnostics

The primary fit uses 201 raw observations from 0 to 2% strain. It retains the intercept and includes negative stresses near the origin. We calculate the residual root-mean-square error as $\sqrt{\sum r_i^2/n}$. Successive samples belong to the same trajectory, so their residual scatter does not estimate variability between independently prepared configurations.

Table S1: Primary-fit diagnostics. n counts fitted observations and b denotes the intercept.

Case	n	b (GPa)	R^2	RMSE (GPa)
Equiatomic	201	-0.0492	0.99621	0.0489
Zr 24%	201	-0.0453	0.99692	0.0421
Zr 28%	201	-0.0432	0.99566	0.0481
Hf 16%	201	-0.0125	0.99640	0.0481
Hf 12%	201	-0.0430	0.99632	0.0490
Ta 16%	201	-0.0829	0.99595	0.0481
Ta 12%	201	-0.0630	0.99417	0.0544
Nb 24%	201	-0.0456	0.99638	0.0508
Nb 28%	201	-0.0272	0.99655	0.0526

Table S2: Fitted stiffness in GPa for five strain intervals. Every fit uses raw stress.

Case	0–1%	0–1.5%	0–2%	0–2.5%	0–3.2%
Equiatomic	137.38	137.05	136.60	135.05	132.05
Zr 24%	130.05	131.28	130.40	128.79	125.91
Zr 28%	125.40	125.49	125.59	124.47	122.46
Hf 16%	136.89	138.10	137.76	137.11	134.36
Hf 12%	139.40	139.14	138.98	138.18	135.96
Ta 16%	132.66	130.89	130.18	128.70	125.73
Ta 12%	123.33	123.18	122.34	120.00	117.35
Nb 24%	148.04	146.86	145.19	143.81	140.40
Nb 28%	156.88	155.33	153.97	152.20	148.64

The small stiffness difference between Hf 16% and the equiatomic case changes sign across the fitting intervals. Its common-window peak remains higher. The two descriptors therefore support different levels of confidence in the composition ordering.

2 Offset construction

For each setting, the fitted elastic line and the offset line share a slope and intercept, with a horizontal displacement of 0.002. The primary extraction averages 21 samples, uses valid centers, searches after the 0–2% fitting interval and requires three consecutive samples at or below the offset line. The crossing lies on the linearly interpolated averaged curve.

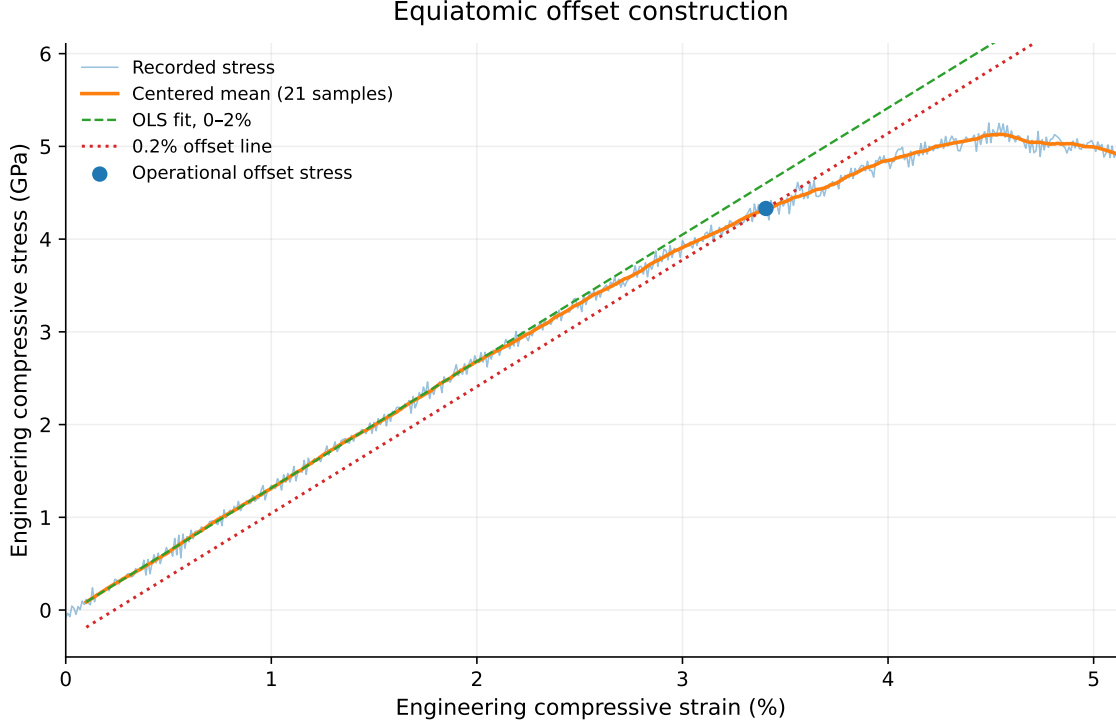


Figure S1: Operational 0.2% offset construction for the equiatomic record. The initial fit and primary peak use raw stress.

We combine five upper fit limits, four centered-average lengths and three persistence counts: 60 settings per curve and 540 curve–setting evaluations. An average length of one retains raw stress. Every setting uses the same 0.002 offset strain and the same 0–10% comparison interval.

Table S3: Operational offset-stress ranges over admissible processing settings.

Case	min σ^{op} (GPa)	max σ^{op} (GPa)	Valid settings
Equiatomic	3.98	4.61	60/60
Zr 24%	3.72	4.44	60/60
Zr 28%	3.71	4.48	60/60
Hf 16%	4.24	4.92	60/60
Hf 12%	4.33	4.96	60/60
Ta 16%	3.10	4.25	60/60
Ta 12%	3.43	4.07	57/60
Nb 24%	3.98	5.19	60/60
Nb 28%	4.35	5.45	60/60

For Ta 12%, all three settings that combine a 3.2% fit limit with raw stress cross the offset line inside the fit interval. The analysis assigns these cases the status `fit_interval_exceeds_offset` and leaves their offset-stress fields empty.

3 Paired comparisons

Every relative difference uses the equiatomic curve processed with the same fit limit, average length and persistence as the modified composition. Averaging and persistence do not change the stiffness fit, so several settings share the same slope.

Table S4: Primary relative changes from the equiatomic record.

Case	ΔE (%)	$\Delta\sigma_{0.2}^{\text{op}}$ (%)	$\Delta\sigma_{\text{p}}$ (%)
Zr 24%	-4.54	-2.07	-4.70
Zr 28%	-8.06	-3.08	-5.44
Hf 16%	+0.85	+6.26	+7.50
Hf 12%	+1.74	+8.91	+10.47
Ta 16%	-4.70	-9.46	-10.70
Ta 12%	-10.44	-13.37	-20.59
Nb 24%	+6.29	+8.97	+10.01
Nb 28%	+12.72	+13.74	+21.56

Table S5: Relative ranges under matched settings. The last column uses each setting’s averaging length, including raw stress; the article’s primary peak always uses raw stress.

Case	ΔE (%)	$\Delta\sigma^{\text{op}}$ (%)	$\Delta\sigma_{\text{p,avg}}$ (%)
Zr 24%	[-5.33, -4.21]	[-7.94, +3.01]	[-4.70, -3.80]
Zr 28%	[-8.72, -7.26]	[-10.22, +1.91]	[-5.58, -4.91]
Hf 16%	[-0.35, +1.75]	[+1.37, +13.93]	[+7.50, +8.16]
Hf 12%	[+1.47, +2.96]	[+2.98, +14.35]	[+10.46, +10.87]
Ta 16%	[-4.79, -3.43]	[-22.42, -5.91]	[-12.50, -10.70]
Ta 12%	[-11.15, -10.12]	[-18.60, -9.76]	[-20.74, -20.46]
Nb 24%	[+6.29, +7.76]	[-0.42, +14.41]	[+10.01, +10.44]
Nb 28%	[+12.57, +14.19]	[+7.57, +19.92]	[+21.56, +22.31]

Zr and Nb 24% offset-stress differences change sign under some settings. Nb 28% retains a higher offset stress across all 60 matched comparisons. Independent configurations, cell sizes and loading rates address separate sources of variation.

4 Complete-record maxima and coverage

Table S6: Complete-record maxima. N counts stress-strain pairs. The final strain determines each maximum’s loading domain.

Case	N	ε_f (%)	$\max \sigma$ (GPa)	$\varepsilon(\max \sigma)$ (%)
Equiatomic	1001	10	5.25	4.49
Zr 24%	2001	20	17.17	19.24
Zr 28%	1001	10	4.97	4.69
Hf 16%	2001	20	15.36	18.62
Hf 12%	2001	20	14.84	18.41
Ta 16%	2001	20	17.93	19.29
Ta 12%	2001	20	16.51	19.20
Nb 24%	2001	20	17.74	19.25
Nb 28%	2001	20	15.68	19.00

The equiatomic and Zr-28% curves end at 10% strain; the other seven reach 20%. The extended records exhibit a second stress rise and higher later maxima. The stress records identify that feature, while the underlying mechanism requires structural and thermodynamic observables.

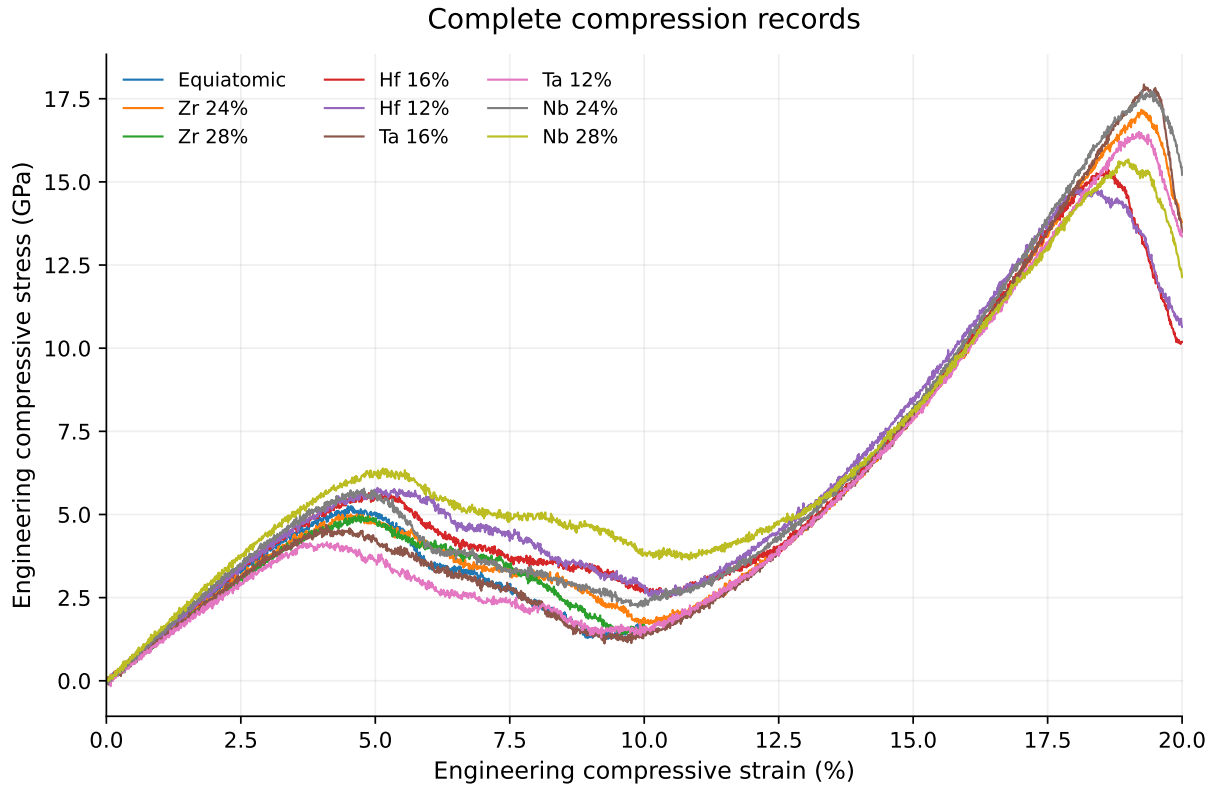


Figure S2: Complete compression records at their recorded strain coverage.

Data and calculations

`results/analysis.json` contains full diagnostics and offset candidates. `results/sensitivity.csv` lists every processing setting and its status. `configs/campaign.json` maps compositions to input files, strain coverage and checksums. Run `python -m hea_md verify` to recalculate the analysis and check the JSON and four CSV exports. `tools/decimal_check.py` independently checks all 540 evaluations using 50-digit decimal arithmetic.

5 Individual composition paths

Each comparison uses raw stress in the common 0–10% interval. The equiatomic curve appears in all four views. The complementary four atomic fractions change equally along each path.

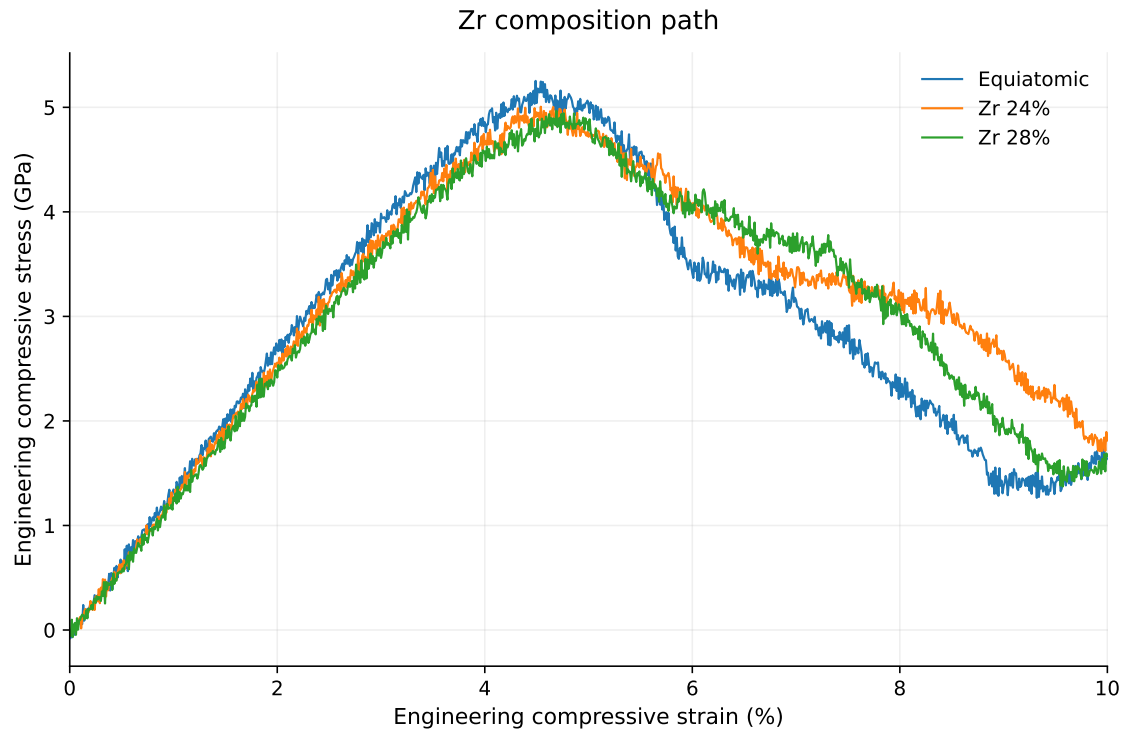


Figure S3: Zr enrichment from 20% to 24% and 28%.

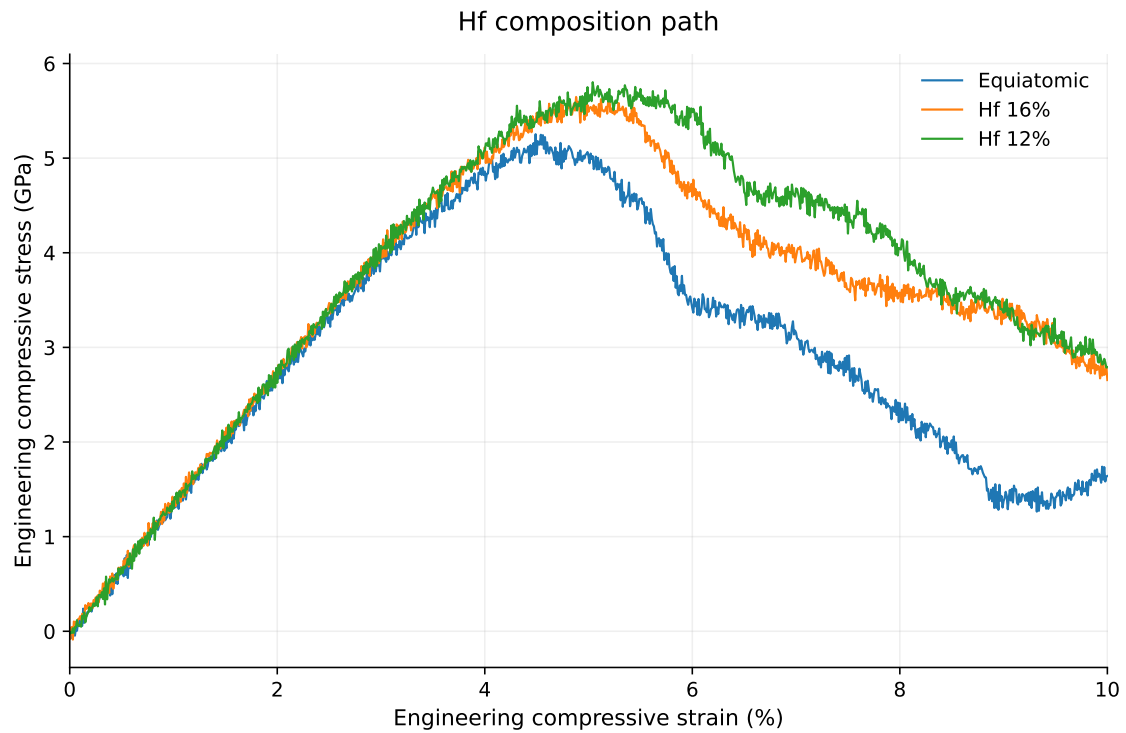


Figure S4: Hf reduction from 20% to 16% and 12%.

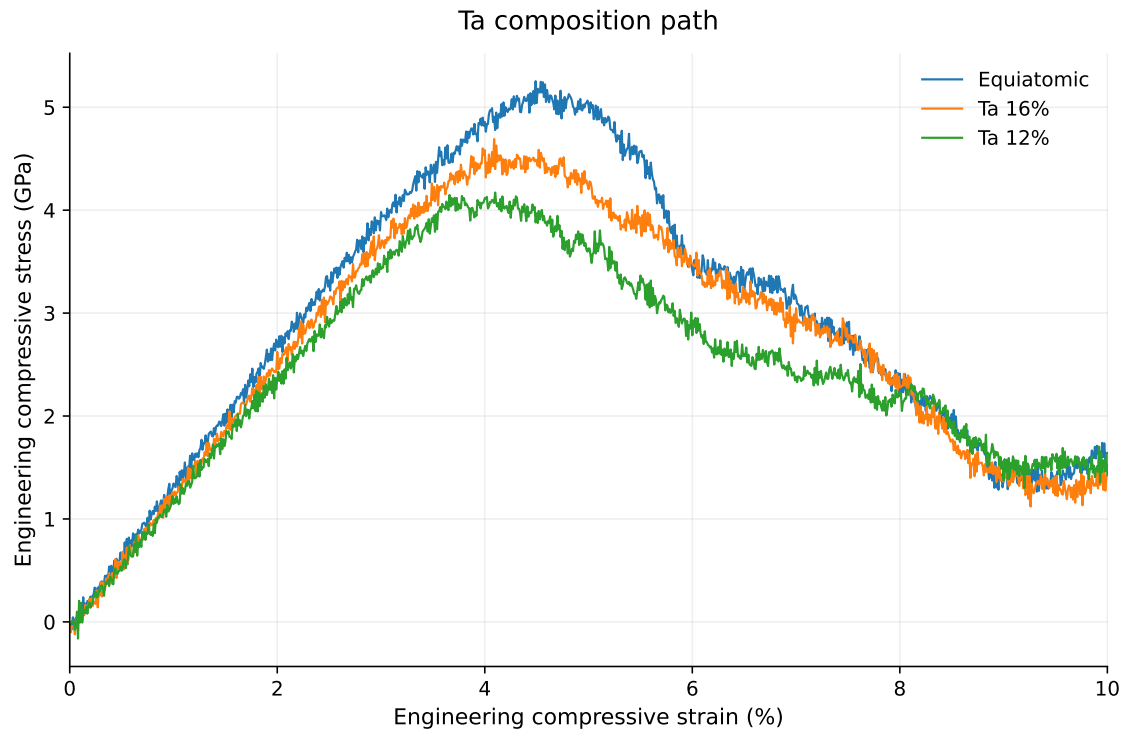


Figure S5: Ta reduction from 20% to 16% and 12%.

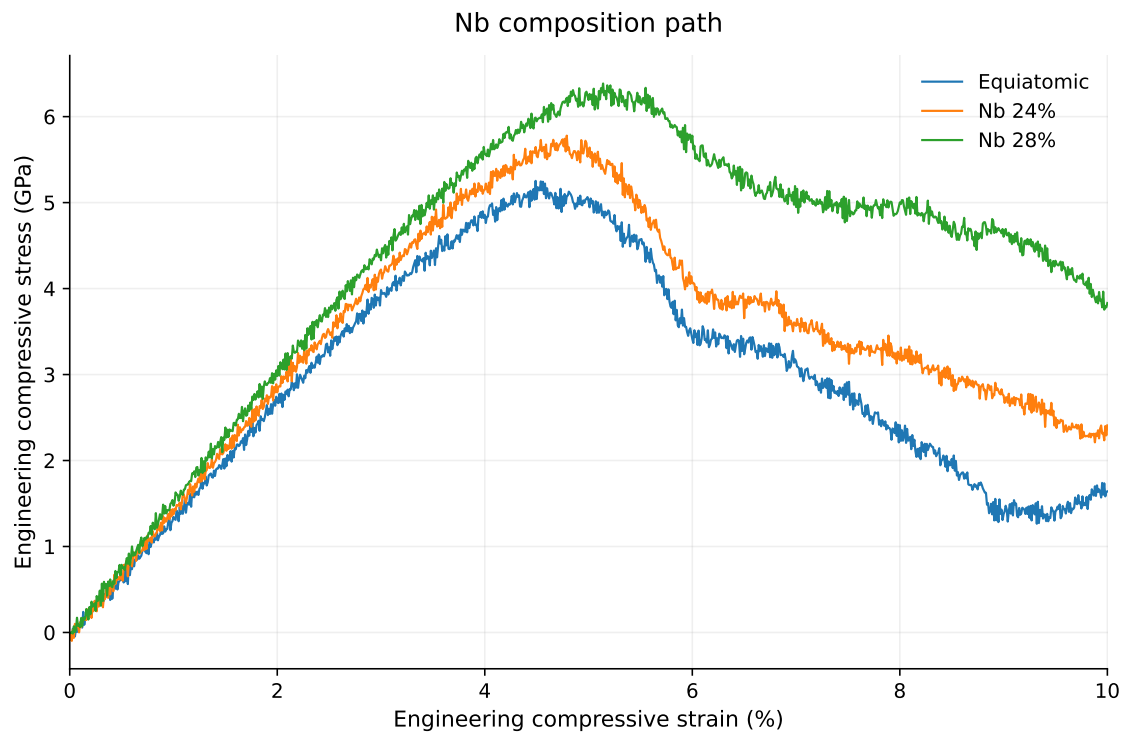


Figure S6: Nb enrichment from 20% to 24% and 28%.